J03 AND J04: FINITE TENSOR ALGEBRA OF GRID COMPLEXES AND THE ITERATED SEAM COMPARISON

Abstract. We prove the statements of jobs J03 and J04 of the dependency audit: the finite tensor algebra of native grid complexes (records M005, M609, M610, M615) and the comparison between the iterated seam grid and the finite tensor of its factors (records M608, M612). All maps are filtered chain maps with shifts $(0,0)$, and each comparison carries the canonical class of its source to the canonical class of its target. The proofs are formalized in Lean, and Section 4 lists the corresponding declarations. As in the record of M609 in the audit, the chain map in M609 is the restriction of one Λ-linear map of the underlying modules, and the constructed map has the same property. Remark 2.9 explains why this hypothesis is needed for the construction.

1. Setting

Throughout, K is a field of characteristic two and $\Lambda = K[T, T^{-1}]$. All homology classes concerning canonical classes are taken with $K = \mathbb{F}_2$.

1.1. **Native grid complexes.** For a grid diagram G of size n with set of states $S(G)$, the module $C(G)$ is the free Λ-module with basis $S(G)$, and $\partial_G \colon C(G) \to C(G)$ is the Λ-linear full rectangle differential in which every O variable is specialized to T (Lean: `laurentDifferential`). A monomial $T^k x$ with $x \in S(G)$ has Maslov degree $M(x) - 2k$ and filtration coordinates $(-k,\, A(x) - k)$, where M and A are the Maslov and Alexander gradings of the grid. For $r \in \mathbb{Z}$ let $C_r(G) \subseteq C(G)$ be the K-span of the monomials of Maslov degree r, and for $a, b \in \mathbb{R}$ let $L_{a,b}(G) \subseteq C(G)$ be the K-submodule of chains all of whose monomials $T^k x$ satisfy $-k \le a$ and $A(x) - k \le b$. The *native complex* $\mathcal{C}(G)$ is the $\mathbb{Z}$-graded complex of K-vector spaces $(C_r(G), \partial_G)$ with the bifiltration $F_{a,b}\mathcal{C}_r(G) = C_r(G) \cap L_{a,b}(G)$ (Lean: `laurentMaslovComplex`, `laurentMaslovBifiltration`). A chain map φ between two bifiltered complexes has *shifts* (s_1, s_2) if $\varphi(F_{a,b}) \subseteq F_{a+s_1, b+s_2}$ for all a, b (Lean: `Bifiltration.Shifts`).

1.2. **Canonical classes.** For a grid diagram G of size at least two, $\xi_G \in H_0(\mathcal{C}(G))$ denotes the normalized class of the project (Lean: `gridCanonicalClass`). For every choice of O variable, it is the unique nonzero class in the image of the corresponding polynomial homology, and c_G denotes the cycle of $\mathcal{C}_0(G)$ with nonnegative exponents chosen to represent it (Lean: `gridCanonicalCycle`). We use the following proved fact.

Proposition 1.1 (`gridCanonicalClass_homotopyEquiv`)**.** *Let G and H be grid diagrams of sizes at least two, and let e be a homotopy equivalence from $\mathcal{C}(G)$ to $\mathcal{C}(H)$ over $\mathbb{F}_2$ whose forward map has shifts $(0, s)$ for some real s. Then $e_*(\xi_G) = \xi_H$.*

1.3. **Finite and binary tensors.** For a finite index type ι and grid diagrams $(G_i)_{i \in \iota}$, put $C(G_\bullet) = \bigotimes_{\Lambda, i \in \iota} C(G_i)$ with the differential $\partial = \sum_{i \in \iota} \partial_i$, where ∂_i applies ∂_{G_i} to the ith factor. Since K has characteristic two, no signs occur and $\partial^2 = 0$. The piece $C_r(G_\bullet)$ is the K-span of the elements $T^j \otimes_i x_i$ with $x_i \in C_{p_i}(G_i)$ and $\sum_i p_i - 2j = r$, and the level $L_{A,B}(G_\bullet)$ is the K-span of the elements $T^j \otimes_i x_i$ with $x_i \in L_{a_i, b_i}(G_i)$, $\sum_i a_i - j \le A$ and $\sum_i b_i - j \le B$. These define the native finite tensor complex $\mathcal{C}(G_\bullet)$ and its bifiltration (Lean: `laurentFiniteMaslovComplex`, `laurentFiniteMaslovBifiltration`). If every G_i has size at least two, the *canonical tensor class* is $\xi_{G_\bullet} = [\otimes_i c_{G_i}] \in H_0(\mathcal{C}(G_\bullet))$ (Lean: `gridCanonicalFiniteTensorClass`). The binary tensor

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$\mathcal{C}(G \otimes H)$ is defined in the same way from $C(G) \otimes_\Lambda C(H)$, and its canonical class is $\xi_{G\otimes H} = [c_G \otimes c_H]$ (Lean: `laurentTensorMaslovComplex`, `gridCanonicalBinaryTensorClass`).

1.4. **Maps induced on pieces.** Each of the three complexes above has chain groups equal to homogeneous K-submodules of one underlying module, a differential induced by one underlying map, and filtration levels induced by one family of underlying submodules (Lean: `PieceComplexData`).

Lemma 1.2 (`PieceComplexData.hom`, `PieceComplexData.hom_shifts`)**.** *Let $\mathcal{C}$ and $\mathcal{C}'$ be complexes of this form with underlying modules M and M', and let $f\colon M \to M'$ be a K-linear map that maps each piece of degree r into the piece of degree r and satisfies $f \circ \partial = \partial' \circ f$. Then the restrictions of f form a chain map $f_\#\colon \mathcal{C} \to \mathcal{C}'$. If moreover $f(L_{a,b}) \subseteq L'_{a,b}$ for all a, b, then $f_\#$ has shifts $(0,0)$. If f is a linear equivalence whose inverse has the same properties, then $f_\#$ is an isomorphism of complexes.*

Proof. The restriction $C_{r+1} \to C'_{r+1}$ composed with ∂' equals ∂ composed with the restriction $C_r \to C'_r$ by $f \circ \partial = \partial' \circ f$. The statement on levels is immediate from the definition of the bifiltrations, and the restrictions of f^{-1} are inverse to those of f. □

2. Finite tensor algebra (J03)

2.1. **The singleton tensor (M005, alias M611).**

Proposition 2.1. *Let ι have exactly one element i_0 and let (G_i) be grid diagrams. The map $c \mapsto \otimes_i c$ induces an isomorphism $\mathcal{C}(G_{i_0}) \cong \mathcal{C}(G_\bullet)$ with shifts $(0,0)$ in both directions. If G_{i_0} has size at least two, the isomorphism maps $\xi_{G_{i_0}}$ to $\xi_{G_\bullet}$.*

Proof. The map $c \mapsto \otimes_i c$ is the inverse of the Λ-linear equivalence $\bigotimes_{i\in\iota} C(G_i) \to C(G_{i_0})$ of Mathlib (`PiTensorProduct.subsingletonEquiv`), which sends $T^j \otimes_i x_i$ to $T^j x_{i_0}$. Since $\sum_i p_i = p_{i_0}$ and $\sum_i a_i = a_{i_0}$, it maps $C_r(G_\bullet)$ into $C_r(G_{i_0})$ and $L_{A,B}(G_\bullet)$ into $L_{A,B}(G_{i_0})$, because multiplication by T^j lowers both coordinates by j. Its inverse maps $C_r(G_{i_0})$ and $L_{A,B}(G_{i_0})$ into the corresponding submodules, taking $j = 0$. The equivalence intertwines ∂ and $\partial_{G_{i_0}}$ by `laurentFiniteDifferential_subsingleton`. Lemma 1.2 gives the isomorphism and the shifts. The forward map sends the cycle $c_{G_{i_0}}$ to $\otimes_i c_{G_i}$, hence $\xi_{G_{i_0}}$ to $\xi_{G_\bullet}$. □

2.2. **Reindexing (M610).**

Proposition 2.2. *Let $e\colon \iota \to \iota'$ be a bijection of finite index types and let $(G_i)_{i\in\iota}$ be grid diagrams. Reindexing induces an isomorphism $\mathcal{C}(G_\bullet) \cong \mathcal{C}(G_{e^{-1}(\bullet)})$ with shifts $(0,0)$ in both directions. If every G_i has size at least two, it maps $\xi_{G_\bullet}$ to $\xi_{G_{e^{-1}(\bullet)}}$.*

Proof. Let R be the Λ-linear equivalence of Mathlib (`PiTensorProduct.reindex`), which sends $T^j \otimes_i x_i$ to $T^j \otimes_{i'} x_{e^{-1}(i')}$. Since $\sum_{i'} p_{e^{-1}(i')} = \sum_i p_i$, the map R sends generators of $C_r(G_\bullet)$ to generators of $C_r(G_{e^{-1}(\bullet)})$, and the same holds for levels. Every pure tensor $\otimes_{i'} y_{i'}$ of the reindexed family equals $R(\otimes_i x_i)$ for the family $x_{e^{-1}(i')} = y_{i'}$, so R^{-1} maps generators to generators as well. For the differential, $R(\partial \otimes_i x_i) = \sum_i R(\otimes_k \tilde{x}^{(i)}_k)$, where $\tilde{x}^{(i)}$ replaces x_i by $\partial_{G_i} x_i$. Substituting $i = e^{-1}(i')$ identifies this sum with $\partial R(\otimes_i x_i)$. Lemma 1.2 gives the isomorphism and the shifts, and R sends $\otimes_i c_{G_i}$ to $\otimes_{i'} c_{G_{e^{-1}(i')}}$. □

If two families of sized grid diagrams are equal as families of pairs (size, diagram), their finite tensor complexes are equal, and the identification preserves the bifiltrations and the canonical tensor classes (Lean: `laurentFiniteMaslovFamilyIso`). This is used for the transport along size equalities in Section 3.

2.3. **Powers of the cyclic row translation (M615).**

Proposition 2.3. *Let G be a grid diagram of size n and $k \geq 0$. Relabeling rows by σ^k, where σ is the cyclic translation of $\{0, \dots, n-1\}$, induces an isomorphism $\mathcal{C}(G) \cong \mathcal{C}(G^{\sigma^k})$ with shifts $(0,0)$ in both directions. If $n \geq 2$, it maps ξ_G to $\xi_{G^{\sigma^k}}$, and its inverse maps $\xi_{G^{\sigma^k}}$ to ξ_G.*

Proof. For $k = 1$, the relabeling preserves the full Laurent coefficients, the Maslov degrees and the Alexander gradings (`laurentCoefficient_relabelRows_finRotate`, `laurentMaslovDegree_relabelRows_finRotate`, `alexander_relabelRows_finRotate`). Since relabeling by σ^{k+1} is relabeling by σ^k followed by relabeling by σ, induction on k gives the three invariances for σ^k. The relabeling $C(G) \to C(G^{\sigma^k})$ of states is Λ-linear, intertwines the differentials and preserves and reflects pieces and levels. Lemma 1.2 gives the isomorphism and the shifts in both directions, and Proposition 1.1 gives the statements on classes. □

2.4. **Insertion of one factor and reassociation (M609).** Let κ be a finite index type presented as the image of an injective map $\iota \to \kappa$ together with one further index $*$ (Lean: `FactorInsertion`). The examples are $\kappa = \iota \sqcup \{*\}$ realized as $\mathrm{Option}\,\iota$, and $\kappa = \{0, \dots, N\}$ with $\iota = \{0, \dots, N-1\}$ and $* = N$. For grid diagrams $(G_k)_{k \in \kappa}$, write G_ι for the restricted family. The assignment $(\otimes_i x_i,\, y) \mapsto \otimes_k z_k$, where $z_i = x_i$ and $z_* = y$, is multilinear in the x_i and linear in y. By the universal property it defines a Λ-bilinear map

$$\mathrm{ins}\colon C(G_\iota) \times C(G_*) \to C(G_\bullet).$$

Lemma 2.4 (`insertLinear_mem_maslovPiece`, `insertLinear_mem_bifiltrationLevel`, `laurentFiniteDifferential_insertLinear`)**.** *For $z \in C_r(G_\iota)$ and $y \in C_q(G_*)$ we have $\mathrm{ins}(z,y) \in C_{r+q}(G_\bullet)$. For $z \in L_{A,B}(G_\iota)$ and $y \in L_{c,d}(G_*)$ we have $\mathrm{ins}(z,y) \in L_{A+c,B+d}(G_\bullet)$. For all z and y,*

$$\partial\, \mathrm{ins}(z,y) = \mathrm{ins}(\partial z, y) + \mathrm{ins}(z, \partial_{G_*} y).$$

Proof. On generators, $\mathrm{ins}(T^j \otimes_i x_i, y) = T^j \otimes_k z_k$, and the degrees and coordinates of the z_k are those of the x_i together with those of y. Their sums are therefore $\sum_i p_i + q$, $\sum_i a_i + c$ and $\sum_i b_i + d$. For the differential, the sum $\sum_{k \in \kappa} \partial_k$ splits into the terms k in the image of ι, which give $\mathrm{ins}(\partial z, y)$, and the term $k = *$, which gives $\mathrm{ins}(z, \partial_{G_*} y)$. Both sides are Λ-linear in z, so the identity on pure tensors suffices. □

Definition 2.5 (`LaurentGridFiniteMap`)**.** Let T be a grid diagram and $(D_i)_{i \in \iota}$ a family of grid diagrams. A *filtered Λ-linear map* $f\colon C(T) \to C(D_\bullet)$ is a Λ-linear map that maps $C_r(T)$ into $C_r(D_\bullet)$, satisfies $f \circ \partial_T = \partial \circ f$, and maps $L_{a,b}(T)$ into $L_{a,b}(D_\bullet)$ for all r, a, b. By Lemma 1.2 it induces a chain map $f_\#\colon \mathcal{C}(T) \to \mathcal{C}(D_\bullet)$ with shifts $(0,0)$.

Definition 2.6 (`LaurentGridInduced`, `LaurentTensorInduced`)**.** A chain map $\theta\colon \mathcal{C}(T) \to \mathcal{C}(D_\bullet)$ is *Λ-induced* if there is a Λ-linear map $f\colon C(T) \to C(D_\bullet)$ whose restriction to $C_r(T)$ is θ_r for every r. The same definition applies to chain maps from a binary tensor complex $\mathcal{C}(T \otimes H)$, with f defined on $C(T) \otimes_\Lambda C(H)$.

Lemma 2.7 (`LaurentGridInduced.exists_laurentGridFiniteMap`)**.** *Let $\theta\colon \mathcal{C}(T) \to \mathcal{C}(D_\bullet)$ be a Λ-induced chain map with shifts $(0,0)$, induced by f. Then f is a filtered Λ-linear map and $f_\# = \theta$.*

Proof. For $c \in C_r(T)$, the element $f(c) = \theta_r(c)$ lies in $C_r(D_\bullet)$. For a state x, the basis vector x lies in $C_{M(x)}(T)$, and the chain map identity for θ in degree $M(x)$ gives $f(\partial_T x) = \partial f(x)$. Both $f \circ \partial_T$ and $\partial \circ f$ are Λ-linear and the states form a Λ-basis of $C(T)$, hence $f \circ \partial_T = \partial \circ f$. A chain $c \in L_{a,b}(T)$ is the K-linear combination of its monomials $T^k x$, and each of these lies in $L_{a,b}(T)$, because the level is defined by a condition on each monomial. Each monomial is homogeneous, so the shifts $(0,0)$ of θ give $f(T^k x) \in L_{a,b}(D_\bullet)$. Summing gives $f(c) \in L_{a,b}(D_\bullet)$. Finally $f_\# = \theta$, because both restrict f. □

Theorem 2.8 (M609)**.** *Let T and H be grid diagrams of sizes at least two, let $(D_i)_{i\in\iota}$ be grid diagrams of sizes at least two, and let D' be the family over* Option ι *with $D'_* = H$ and $D'_i = D_i$. Let $\theta\colon \mathcal{C}(T) \to \mathcal{C}(D_\bullet)$ be a Λ-induced chain map with shifts $(0,0)$ and $\theta_*(\xi_T) = \xi_{D_\bullet}$. Then there is a Λ-induced chain map $\psi\colon \mathcal{C}(T \otimes H) \to \mathcal{C}(D'_\bullet)$ with shifts $(0,0)$ and $\psi_*(\xi_{T\otimes H}) = \xi_{D'_\bullet}$.*

Proof. By Lemma 2.7, $\theta = f_\#$ for a filtered Λ-linear map f. More generally, let κ be presented as above and let $f\colon C(T) \to C(G_\iota)$ be a filtered Λ-linear map. The map $(c, y) \mapsto \mathrm{ins}(f(c), y)$ is Λ-bilinear, so it defines a Λ-linear map $\Psi_f\colon C(T) \otimes_\Lambda C(G_*) \to C(G_\bullet)$ with $\Psi_f(c \otimes y) = \mathrm{ins}(f(c), y)$. Lemma 2.4 shows that Ψ_f maps the generators $c \otimes y$ of the binary pieces and levels into the corresponding pieces and levels, using monotonicity of the levels. For the differential,

$$\Psi_f(\partial c \otimes y + c \otimes \partial y) = \mathrm{ins}(\partial f(c), y) + \mathrm{ins}(f(c), \partial y) = \partial\, \Psi_f(c \otimes y).$$

By Lemma 1.2, $\psi = (\Psi_f)_\#$ has shifts $(0,0)$, and ψ is Λ-induced by Ψ_f (Lean: `LaurentGridFinite eMap.insertTensor`, `LaurentTensorFiniteMap.laurentTensorInduced`).

For the classes, let y be a cycle of degree zero in $\mathcal{C}(G_*)$. The map $S_y = \mathrm{ins}(-, y)$ is a chain map $\mathcal{C}(G_\iota) \to \mathcal{C}(G_\bullet)$ by Lemma 2.4, because $\partial y = 0$ (Lean: `insertCycleHom`). For a cycle c of degree zero in $\mathcal{C}(T)$, the cycles $\Psi_f(c \otimes y)$ and $S_y(f(c))$ coincide, so $\psi_*[c \otimes y] = S_{y*}(f_{\#*}[c])$. With $c = c_T$ and $y = c_{G_*}$ this gives

$$\psi_*(\xi_{T\otimes G_*}) = S_{y*}(\xi_{G_\iota}) = [\mathrm{ins}(\otimes_i c_{G_i}, c_{G_*})] = [\otimes_k c_{G_k}] = \xi_{G_\bullet}.$$

The case $\kappa = $ Option ι and $G = D'$ is the theorem. □

Remark 2.9. The proposal for M609 cited in the job listing of J03 allows an arbitrary chain map $\theta\colon \mathcal{C}(T) \to \mathcal{C}(D_\bullet)$ of complexes of K-vector spaces. The formula $c \otimes y \mapsto \theta(c) \otimes y$ is well defined on the balanced tensor product $C(T) \otimes_\Lambda C(H)$ only if it respects the relation $(Tc) \otimes y = c \otimes (Ty)$, which requires $\theta(Tc) = T\theta(c)$. The record of M609 in the dependency audit adds the hypothesis that θ is Λ-induced and the same conclusion for ψ, and Theorem 2.8 is that statement. The final comparison map of the route is a composite $\theta_{\mathrm{tr}} \circ \varphi$, where φ comes from the cobordism and the iterated comparison θ_{tr} of Section 3 is applied last. Composition requires no linearity of φ, so no other map of the route needs to be Λ-induced. We do not know whether the proposal for an arbitrary chain map holds, and it is not needed.

3. The iterated seam comparison (J04)

3.1. **The binary seam (M608).** For grid diagrams G of size $n+1$ and H of size $m+1$ with $X_G(n) = n$ and $X_H(0) = 0$, the shared corner seam grid $\mathrm{Seam}(G, H)$ of size $n+m+1$ is `connectedSumSeamGrid`. Assume also $O_H(m) = 0$. The seam contraction gives a Λ-linear projection $\pi\colon C(\mathrm{Seam}(G,H)) \to C(G) \otimes_\Lambda C(H)$, the corrected projection followed by the inverse tensor coordinates (Lean: `seamTensorProjection`). It maps levels into levels (`seamFullContraction_proj_mem_bifiltration_of_last_0`, `seamTensorCoordinatesEquiv_symm_mem_bifiltration`).

Lemma 3.1 (M608, `seamParity_of_positivePaths`)**.** *If the seam positive path comparisons hold for the pair of sizes (n, m), then the parity identity used as the hypothesis of* `seamNativeTensorHomotopyEquiv` *holds for every such pair G, H.*

Proof. The comparisons are equivalent to the equality of the retained seam differential with the tensor coordinate differential over $\mathbb{F}_2$ (`seamRetainedDifferential_eq_tensorCoordinate_iff_positive_paths`), and this equality is equivalent to the parity identity (`seamRetainedDifferential_eq_tensorCoordinate_iff_evalOne`). □

Under the parity identity, π commutes with the differentials and maps the piece of degree r into the piece of degree r, and $\pi_\#$ is the forward map of the homotopy equivalence `seamNativeTensorHomotopyEquiv`. It satisfies $\pi_{\#*}(\xi_{\mathrm{Seam}(G,H)}) = \xi_{G\otimes H}$ by `seamNativeTensor_hom_canonicalClass`.

The comparisons hold for all sizes by M601 (`seamPositivePathComparisons`), so Lemma 3.1 applies without hypothesis.

3.2. **Prepared families and the iterated seam grid.**

Definition 3.2 (`SeamPrepared`)**.** A family $(D_j)_{0\le j<N}$ of grid diagrams of sizes s_j is *prepared* if every s_j is positive, $X_{D_0}(s_0-1)=s_0-1$ when $N>0$, and $X_{D_j}(0)=0$ and $O_{D_j}(s_j-1)=0$ for every $j\neq 0$.

For a grid diagram G of size $n+1$, some power σ^k of the cyclic row translation satisfies $\sigma^k(X_G(n))=n$ (`exists_cyclicRows_send` for $n\ge 1$, and $k=0$ for $n=0$). We fix one such power and put $\mathrm{rot}(G)=G^{\sigma^k}$, so that $X_{\mathrm{rot}(G)}(n)=n$ (Lean: `seamLeftGrid`).

Let $(E_j)_{0\le j\le M}$ be grid diagrams of sizes p_j+1 with $X_{E_j}(0)=0$ for $j\ge 1$. The *seam chain* is defined by

$$T_0=E_0,\qquad T_{k+1}=\mathrm{Seam}(\mathrm{rot}(T_k),E_{k+1})\quad(0\le k<M)$$

(Lean: `seamChainGrid`). By induction T_k has size $\sum_{j\le k}p_j+1$. For a prepared family $(D_j)_{0\le j<N}$ with $N=M+1$, put $p_j=s_j-1$ and let E_j be D_j viewed as a grid of size p_j+1. The *iterated seam grid* of D is T_M, transported to the size $\sum_j s_j-(N-1)=\sum_j p_j+1$ (Lean: `iteratedSeamGrid`, `iteratedSeamSize`, `iteratedSeamSize_succ`). For $N=0$ it is the grid of size zero.

Lemma 3.3 (`two_le_iteratedSeamSize`)**.** *If $N>0$ and $s_j\ge 2$ for all j, then $\sum_j s_j-(N-1)\ge 2$.*

Proof. We have $\sum_j s_j\ge 2N$, hence $\sum_j s_j-(N-1)\ge N+1\ge 2$. □

Lemma 3.4 (`exists_seamPrepared_movesTo`)**.** *Every family (D_j) of grid diagrams of sizes at least two is related factorwise by grid moves to a prepared family of grid diagrams of sizes at least two.*

Proof. For $j=0$, relabel the rows of D_0 by a power of σ that sends $X_{D_0}(s_0-1)$ to s_0-1 (`exists_cyclicRows_send`). Cyclic row relabelings are grid moves (`cyclicRows_pow_movesTo`). For $j\neq 0$, `exists_seamPreparedGrid` gives a grid D'_j of size s_j+1 reached by grid moves with $X_{D'_j}(0)=0$ and $O_{D'_j}(s_j)=0$. □

3.3. **The comparison (M612).**

Theorem 3.5 (M612)**.** *Assume the seam positive path comparisons for all pairs of sizes. Let $N>0$ and let $(D_j)_{0\le j<N}$ be a prepared family of grid diagrams of sizes at least two, with iterated seam grid G_{tr}. Then there is a chain map $\theta\colon \mathcal{C}(G_{\mathrm{tr}})\to\mathcal{C}(D_\bullet)$ over $\mathbb{F}_2$ with shifts $(0,0)$ and $\theta_*(\xi_{G_{\mathrm{tr}}})=\xi_{D_\bullet}$. Since the comparisons are proved (M601), the conclusion holds without hypothesis.*

Proof. We first prove, by induction on M, the following statement for the seam chain of grid diagrams $(E_j)_{0\le j\le M}$ of sizes $p_j+1\ge 2$ with $X_{E_j}(0)=0$ and $O_{E_j}(p_j)=0$ for $j\ge 1$: there is a filtered Λ-linear map $f_M\colon C(T_M)\to C(E_\bullet)$ with $f_{M\#*}(\xi_{T_M})=\xi_{E_\bullet}$ (Lean: `exists_seamChain_finiteTensor_map`).

For $M=0$, Proposition 2.1 gives f_0, the inverse singleton equivalence.

For the inductive step, let $f_M\colon C(T_M)\to C(E_{\le M})$ be given for the family $(E_j)_{j\le M}$, and let k be the chosen power with $\mathrm{rot}(T_M)=T_M^{\sigma^k}$.

(1) The inverse row relabeling $C(\mathrm{rot}(T_M))\to C(T_M)$ is a filtered Λ-linear map of grid chains (Proposition 2.3), and its induced chain map is the inverse isomorphism of Proposition 2.3. Hence $g=f_M\circ\sigma^{-k}$ is a filtered Λ-linear map $C(\mathrm{rot}(T_M))\to C(E_{\le M})$, and

$$g_{\#*}(\xi_{\mathrm{rot}(T_M)})=f_{M\#*}(\xi_{T_M})=\xi_{E_{\le M}}$$

by Proposition 2.3 and the induction hypothesis.

(2) Insertion of the factor E_{M+1} into the last position, with $\kappa = \{0, \dots, M+1\}$ and $* = M+1$, gives the map $\Psi_g \colon C(\mathrm{rot}(T_M)) \otimes_\Lambda C(E_{M+1}) \to C(E_{\le M+1})$ of the proof of Theorem 2.8, with $(\Psi_g)_{\#*}(\xi_{\mathrm{rot}(T_M)\otimes E_{M+1}}) = \xi_{E_{\le M+1}}$.

(3) The grid $\mathrm{rot}(T_M)$ satisfies $X_{\mathrm{rot}(T_M)}(s) = s$, where $s+1$ is its size, and E_{M+1} satisfies $X_{E_{M+1}}(0) = 0$ and $O_{E_{M+1}}(p_{M+1}) = 0$. The parity identity holds for this pair by Lemma 3.1, and $T_{M+1} = \mathrm{Seam}(\mathrm{rot}(T_M), E_{M+1})$. Put $f_{M+1} = \Psi_g \circ \pi$. It is a filtered Λ-linear map because π and Ψ_g preserve pieces, levels and differentials. Its induced chain map is $(f_{M+1})_\# = (\Psi_g)_\# \circ \pi_\#$, as recorded in `LaurentTensorFiniteMap.precompSeam_hom`. Therefore

$$f_{M+1\,\#*}(\xi_{T_{M+1}}) = (\Psi_g)_{\#*}(\xi_{\mathrm{rot}(T_M)\otimes E_{M+1}}) = \xi_{E_{\le M+1}}.$$

For the theorem, write $N = M+1$ and let E_j be D_j viewed as a grid of size $(s_j - 1) + 1$. These grids satisfy the hypotheses above because the family is prepared and $s_j \ge 2$. Let f_M be the map just constructed. Put

$$\theta = \tau_D \circ f_{M\#} \circ \tau_{\mathrm{tr}}^{-1},$$

where $\tau_{\mathrm{tr}} \colon \mathcal{C}(T_M) \to \mathcal{C}(G_{\mathrm{tr}})$ is the transport along the size equality and $\tau_D \colon \mathcal{C}(E_\bullet) \to \mathcal{C}(D_\bullet)$ is the transport of families. Both transports are identities after the size equalities are substituted, so they have shifts $(0,0)$, τ_{tr}^{-1} maps $\xi_{G_{\mathrm{tr}}}$ to ξ_{T_M} by Proposition 1.1, and τ_D maps $\xi_{E_\bullet}$ to $\xi_{D_\bullet}$. Shifts add under composition (`Bifiltration.Shifts.comp`), so θ has shifts $(0,0)$ and $\theta_*(\xi_{G_{\mathrm{tr}}}) = \xi_{D_\bullet}$. □

Remark 3.6. The condition $X_{D_0}(s_0 - 1) = s_0 - 1$ in Definition 3.2 is part of the audit's definition and is not used in the proof, because rot is applied at every step. The theorem does not identify G_{tr} with a grid of the connected sum of the knots of the factors. That geometric statement is record M613 and remains open.

4. Correspondence with the Lean declarations

All declarations are in the namespace `OnLeConjecture.KnotFloer.GridFull`. The audits `AuditGridLaurentFiniteTensorAlgebra.lean` and `AuditGridConnectedSumIterated.lean` check that every declaration of the six modules, including internal ones, depends only on `propext`, `Classical.choice` and `Quot.sound`.

Statement	Lean declaration and module
Lemma 1.2	`PieceComplexData.hom`, `PieceComplexData.iso`, `PieceComplexData.hom_shifts` (`GridLaurentFiniteTensorSingleton`)
Proposition 2.1	`laurentFiniteMaslovSubsingletonIso` with `_hom_shifts`, `_inv_shifts`, `_hom_canonicalClass`, `_inv_canonicalClass` (`GridLaurentFiniteTensorSingleton`)
Proposition 2.2	`laurentFiniteMaslovReindexIso` with `_hom_shifts`, `_inv_shifts`, `_hom_canonicalClass`, `_inv_canonicalClass`, and `laurentFiniteMaslovFamilyIso` (`GridLaurentFiniteTensorReindex`)
Proposition 2.3	`laurentMaslovCyclicRowsPowIso` with `_hom_shifts`, `_inv_shifts`, `gridCanonicalClass_cyclicRowsPow`, `gridCanonicalClass_cyclicRowsPow_inv` (`GridFullCyclicRowsPower`)

Lemma 2.4, Definitions 2.5 and 2.6, Lemma 2.7	`insertLinear`, `LaurentGridFiniteMap`, `LaurentTensorFiniteMap`, `LaurentGridInduced`, `LaurentTensorInduced`, `LaurentGridInduced.exists_laurentGridFiniteMap` (`GridLaurentFiniteTensorReassociation`)
Theorem 2.8	`exists_tensorGrid_finiteTensor_map`, `LaurentGridFiniteMap.insertTensor_canonicalClass` (`GridLaurentFiniteTensorReassociation`)
Lemma 3.1	`seamParity_of_positivePaths` (`GridConnectedSumSeamUnconditional`)
Definition 3.2, Lemmas 3.3 and 3.4	`SeamPrepared`, `iteratedSeamSize`, `iteratedSeamGrid`, `two_le_iteratedSeamSize`, `exists_seamPrepared_movesTo` (`GridConnectedSumIterated`)
Theorem 3.5	`exists_seamChain_finiteTensor_map`, `exists_iteratedSeam_finiteTensor_map_of_parity`, `exists_iteratedSeam_finiteTensor_map` (`GridConnectedSumIterated`)

Deviations from the proposed API.

(1) The index types of the isomorphisms of M005, M610 and the family transport are in the universe `Type`. For $K = \mathbb{F}_2$ the native grid complex lives in the universe of `Type`, and a finite tensor over an index type of a higher universe lives in that higher universe, so the proposed isomorphisms do not typecheck for such index types.

(2) M609 carries the hypothesis that θ is Λ-induced and the same conclusion for ψ, as in the record of M609 in the audit. The proposal cited in the job listing of J03 omits both (Remark 2.9). This formulation was approved on 2 November 2026.

(3) In addition to the proposed conditional theorem, the module proves the unconditional version `exists_iteratedSeam_finiteTensor_map`, using M601.